SALES PERFORMANCE DASHBOARD

Power BI Project Documentation & Business Insights

Objective
Build an interactive sales report to evaluate revenue, profit, profit margin, units sold, regional performance, product categories, salespeople, channels, discount levels, and marketing spend.

Tools: Power BI Desktop, Power Query, DAX, CSV dataset

Dataset: large_sales_data.csv — 1,000 rows and 11 columns, based on the project review.

The original public dataset source URL was not provided in the conversation. Add the actual source name, URL, and licence details to the GitHub README; do not guess or invent a source.

1. Project Overview

This project contains three interactive report pages. KPI cards, charts, tables/matrices, and slicers let users compare performance by region, category, salesperson, and sales channel.

Business questions

• What are total revenue, total profit, profit margin, and units sold?

• Which regions contribute most to revenue and profit?

• How do product categories compare in revenue, profit, and unit volume?

• How do salespeople compare across revenue, profit, and units sold?

• How do Online, Retail, and Wholesale channels compare?

• How do discounts and marketing spend vary across segments?

2. Dataset Fields

Field	Analytical purpose
Region	Regional comparison
Salesperson	Sales-team comparison
Revenue	Sales value
Profit	Reported profit
Units_Sold	Sales volume
Customer_Satisfaction	Satisfaction-score comparison
Marketing_Spend	Marketing expenditure
Discount_Percentage	Discount analysis
Product_Category	Category comparison
Sales_Channel	Online/Retail/Wholesale comparison
Order_Processing_Time	Processing-time analysis

3. Step-by-Step Workflow

Step 1 — Import the CSV. In Power BI Desktop, select Home → Get data → Text/CSV, choose large_sales_data.csv, review the preview, and select Transform Data to open Power Query Editor.

Step 2 — Inspect the data. Review column names, sample values, data types, column quality, errors, and blank values. The initial inspection reported no missing values.

Step 3 — Set data types. Use Text for Region, Salesperson, Product_Category, and Sales_Channel. Use appropriate numeric types for Revenue, Profit, Units_Sold, Marketing_Spend, Discount_Percentage, Customer_Satisfaction, and Order_Processing_Time. Confirm each field's unit/scale from source documentation.

Step 4 — Validate data quality. Check unexpected negative values, inconsistent labels, duplicates where a valid unique identifier exists, and implausible business relationships. Investigate unusual values instead of automatically deleting or changing them.

Step 5 — Apply and load. After reviewing Power Query transformations, select Close & Apply to load the table. The shown report is based on one main sales table; a multi-table relationship model was not necessary for the displayed analyses.

Step 6 — Create DAX measures. Measures provide consistent calculations and respond to slicers.

Measure	DAX expression
Total Revenue	Total Revenue = SUM(SalesData[Revenue])
Total Profit	Total Profit = SUM(SalesData[Profit])
Total Units Sold	Total Units Sold = SUM(SalesData[Units_Sold])
Profit Margin %	Profit Margin % = DIVIDE([Total Profit], [Total Revenue], 0)

Measure	DAX expression
Average Satisfaction	Average Customer Satisfaction = AVERAGE(SalesData[Customer_Satisfaction])
Total Marketing Spend	Total Marketing Spend = SUM(SalesData[Marketing_Spend])
Average Discount	Average Discount % = AVERAGE(SalesData[Discount_Percentage])
Average Processing Time	Average Processing Time = AVERAGE(SalesData[Order_Processing_Time])

Format Profit Margin % as Percentage (the screenshot displayed 0.22; this is approximately 22%). Confirm whether Discount_Percentage is stored as percentage points (15 = 15%) or a fraction (0.15 = 15%) before formatting it.

4. Report Pages and Visuals

Page 1 — Executive Sales Overview
KPI cards for Total Revenue, Total Profit, Profit Margin %, and Total Units Sold; profit and revenue by product category; profit and revenue by region; salesperson summary table; slicers for Sales Channel, Region, and Product Category.

Page 2 — Regional & Product Analysis
Revenue contribution by region and category; profit by region; units sold by category; region-by-category matrix; scatter plot comparing average discount with profit margin by region.

Page 3 — Salesperson & Sales Channel Performance
Revenue and profit by salesperson; salesperson performance table; revenue and profit by sales channel; channel-level comparison table; interactive slicers.

5. Top Business Insights

These are descriptive findings visible in the supplied screenshots. Verify the final measures, filters, and business definitions before publishing.

1. Overall performance: The report displays approximately 107.3 million in revenue and 24.0 million in profit. Overall profit margin is approximately 22.4%, subject to confirmation of the currency and profit definition.

2. West is the leading region: West shows approximately 29 million in revenue and 7 million in profit, the highest displayed regional totals. Management could investigate the factors behind its performance before considering whether they apply elsewhere.

3. Regional performance differs: North and West have the largest displayed revenue totals, while South has the smallest. South may warrant further investigation into sales mix, discounting, and operating conditions; the dataset does not establish the cause.

4. Category revenue is balanced: The five categories contribute similar revenue totals, broadly around 20–23 million each. No single category overwhelmingly dominates the displayed revenue mix.

5. Clothing leads in units sold: The category chart shows Clothing at approximately 109K units, followed by Electronics at approximately 103K and Grocery at approximately 97K. Unit volume alone does not prove stronger profitability.

6. Salesperson revenue is relatively close: Displayed salesperson revenue ranges from about 18.1 million to 22.9 million. Compare profit margin and customer satisfaction alongside revenue rather than ranking people on revenue alone.

7. Channel revenue is similar: Online, Retail, and Wholesale each generate approximately 35–36 million in displayed revenue. Online shows about 8.19 million profit and Wholesale about 7.53 million; recheck measures and filters before publication.

8. Discount and margin vary: The regional scatter plot shows differences in average discount and profit margin. This is a useful investigation lead, not proof that discounts cause margin changes.

6. Data Quality Checks & Limitations

A validation check identified 74 rows where reported Profit exceeded Revenue. Confirm the source's profit formula and business rules before correcting those records or interpreting them. This may be a data-definition issue, a characteristic of the generated dataset, or a data-quality problem.

• No date field: month-over-month growth, year-over-year trends, and seasonality cannot be analysed.

• No order ID or customer ID: order counts, repeat-customer behaviour, and customer retention cannot be calculated.

• No individual product names: analysis is limited to product categories.

• Marketing spend can be compared with revenue, but campaign ROI or causal marketing effectiveness cannot be established from these fields alone.

• Confirm currency, discount scale, satisfaction scale, and processing-time units from the source documentation.

• Validate how Marketing_Spend is aggregated before publishing conclusions about expenditure.

7. README Summary

This Power BI project analyses sales performance across regions, product categories, salespeople, and sales channels. It includes KPI measures for revenue, profit, profit margin, and units sold, supported by interactive visuals and slicers.

The report highlights revenue contribution, profitability differences, sales-volume patterns, and areas for further investigation. Data-quality limitations are documented, and observed relationships are treated as descriptive findings rather than causal conclusions.

8. GitHub Publishing Checklist

- Add the verified dataset source name, URL, and licence/permission details to README.md.
- Include the .pbix file if appropriate and ensure it contains no confidential information.
- Add screenshots of all three report pages.
- Place this PDF in a documentation folder and link it from README.md.
- Verify DAX measures, number formats, units, and slicer interactions.
- Document the Profit > Revenue validation issue and dataset limitations.